

▲ Plate 6.5 Salted dried

are made using this principle. Vegetables like sundaikai, cluster beans, bitter gourd and green chillies are preserved by this method. Fish and meat are also sun dried. The common dried fish or karuvadu (local name) is a good example for sun drying.

ii. Drying by Mechanical Driers: Artificial drying involves the passage of hot air with controlled relative humidity over the food to be dried or the passage of the food through such air. Fruits, vegetables, nuts, fish and meat can be successfully preserved by this method. In the dehydration process, artificial drying methods like spray drying, vacuum drying, drum drying and freeze drying are used for drying foods. Although it is expensive when compared to natural sun-drying process, it is very hygienic, rapid and the products obtained are uniform in colour because the temperature and relative humidity can be maintained.

i. Spray drying: Milk and eggs are dried to a powder in spray driers in which the liquid is atomized and sprayed into hot air steam or almost instant drying.
ii. Foam mat drying: Foam mat drying may be used commercially to preserve orange and tomato juice. In this process a small amount of edible foam stabilizer is used. The foam is spread in a thin layer and dried in a steam of hot air. The product gets separated easily into small particles on cooling.
iii. Drying by osmosis: Drying by osmosis results when fish is heavily salted. In this case, the moisture is drawn out from all the cell tissues. The water is then bound with the solute making it unavailable to the microorganisms. In osmotic dehydration of fruits, the method involves the partial dehydration of fruits by osmosis in a concentrated sugar solution or syrup.



▲ Plate 6.6 Spray dried milk powder



▲ Plate 6.7 Dehydrated Foods

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